⚛️ React.js — Complete Revision Course Outline

> **Goal:** One revision PDF covering every major React topic, written in simple English.
> **Target length:** ~80 pages (range 75–85) **Format per topic:** Tagline → Explanation → Code → Key Points → Memory Tip **Delivery:** 6 Markdown files (one per part), merged into one PDF at the end

---

📊 Overview

Part	Title	Topics	Pages
1	Fundamentals	13	16
2	Hooks	10	20
3	Routing & Data	4	9
4	Performance & Quality	7	10
5	State Management & Ecosystem	6	15
6	Modern React & Extras	6	10
	TOTAL	**46**	**≈ 80**

Each part ends with a cheat sheet table (included in its page count).

🧱 Part 1 — Fundamentals `16 pages`

#	Topic	What it covers	Pages
1	What is React	Library vs framework, virtual DOM, reconciliation, SPA, why React	1

#	Topic	What it covers	Pages
2	Setup & Project Structure	Vite, npm, folder structure, `main.jsx`, `App.jsx`, scripts	1
3	JSX	Syntax rules, `{}` expressions, fragments, `className`, self-closing tags	1.5
4	Components	Function components, composition, naming, reusability, export/import	1
5	Props	Passing data, destructuring, default props, `children`, read-only rule	1.5
6	State Basics	What state is, props vs state, re-rendering	1
7	Event Handling	`onClick`, `onChange`, synthetic events, passing arguments, `preventDefault`	1
8	Conditional Rendering	`if`, ternary, `&&`, early return, null rendering	1
9	Lists & Keys	`map()`, why keys matter, index-as-key problem, filtering	1
10	Forms	Controlled vs uncontrolled, multiple inputs, checkbox/select, validation basics	2
11	Lifting State Up	Sharing state between siblings, parent-child communication	1
12	Component Lifecycle	Mount, update, unmount, class vs hooks mapping	1.5
13	Styling	CSS files, CSS modules, inline styles, Tailwind, styled-components	1.5

🪝 Part 2 — Hooks `20 pages`

#	Topic	What it covers	Pages
14	What is a Hook + Rules of Hooks	Why hooks exist, top-level rule, React-functions-only rule, DO/DON'T	1
15	`useState`	Syntax, functional updates, objects/arrays, lazy init, batching, async nature	2.5
16	`useEffect`	Dependency array, cleanup, fetch, timers, common mistakes, infinite loops, Strict Mode	3
17	`useContext`	`createContext`, Provider, consumers, prop drilling, re-render behavior	2
18	`useRef`	DOM access, mutable values, previous value, ref vs state	2
19	`useMemo`	Memoizing values, dependencies, when NOT to use	1.5
20	`useCallback`	Memoizing functions, pairing with `React.memo`, vs `useMemo`	1.5
21	`useReducer`	Reducer, actions, dispatch, vs `useState`, Redux connection	2
22	Custom Hooks	Naming, extracting logic, `useFetch`, `useLocalStorage`, `useDebounce`, `useToggle`	2.5
23	Other Hooks	`useLayoutEffect`, `useId`, `useTransition`, `useDeferredValue`, `useImperativeHandle`	2

🧭 Part 3 — Routing & Data `9 pages`

#	Topic	What it covers	Pages
24	React Router	`BrowserRouter`, `Routes`, `Link` / `NavLink`, params, nested routes, `Outlet`, `useNavigate`, `useParams`, 404	3.5
25	Protected Routes	Auth guard component, redirects, role-based routes	1.5

#	Topic	What it covers	Pages
26	Data Fetching	`fetch` vs Axios, loading/error states, async/await, cancelling requests	2.5
27	Context API Patterns	Auth context, theme context, custom provider, avoiding re-render traps	1.5

🚀 Part 4 — Performance & Quality `10 pages`

#	Topic	What it covers	Pages
28	`React.memo`	How it works, shallow comparison, when it helps	1
29	Code Splitting	`React.lazy`, `Suspense`, route-based splitting	1.5
30	Error Boundaries	Class-based boundary, fallback UI, what they can't catch	1
31	Portals	`createPortal`, modals, tooltips	1
32	Refs & `forwardRef`	Passing refs to children, exposing DOM nodes	1.5
33	HOCs & Render Props	Patterns, examples, why hooks replaced them	1.5
34	Performance Optimization	Unnecessary re-renders, key usage, virtualization, React DevTools Profiler	2.5

🗄️ Part 5 — State Management & Ecosystem `15 pages`

#	Topic	What it covers	Pages
35	Redux Toolkit	Store, slices, reducers, `dispatch`, `useSelector`, `createAsyncThunk`, flow diagram	4
36	Zustand	Create store, use in components, vs Redux	1.5

#	Topic	What it covers	Pages
37	TanStack Query	`useQuery`, `useMutation`, caching, refetching, query keys	2.5
38	React Hook Form	`register`, `handleSubmit`, errors, validation, Zod basics	2
39	Authentication Flow	JWT, login/logout, token storage, protected API calls	2.5
40	Connecting to Backend (MERN)	Axios setup, CORS, env variables, full request cycle	2.5

✨ Part 6 — Modern React & Extras `10 pages`

#	Topic	What it covers	Pages
41	React 19 Features	Actions, `use`, `useOptimistic`, `useFormStatus`, `useActionState`, ref as prop	2
42	Server Components	Concept, client vs server components, when to use (overview)	1.5
43	TypeScript with React	Typing props, state, events, hooks	2
44	Testing	Jest, React Testing Library, rendering, queries, user events	2
45	Deployment	Build process, Vercel, Netlify, env variables	1
46	Interview Questions	Top 25–30 questions with short answers	1.5

🗂️ Delivery Plan

Response	File	Parts
1	`React_Part1_Fundamentals.md`	Topics 1–13
2	`React_Part2_Hooks.md`	Topics 14–23
3	`React_Part3_Routing_Data.md`	Topics 24–27
4	`React_Part4_Performance_Quality.md`	Topics 28–34
5	`React_Part5_State_Ecosystem.md`	Topics 35–40
6	`React_Part6_Modern_Extras.md`	Topics 41–46

✅ Topic Template (used for every topic)

```
## Topic Name
> One-line tagline

Explanation (2-3 simple lines)

Code example (commented)

KEY POINTS
- 6 to 10 short, explainable bullets

💡 Memory tip: one-line analogy or rule of thumb
```